Determinants of Malnutrition among under-five children in Kenya

Application of the Modified Poisson Regression Approach

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Introduction

- Malnutrition is defined as imbalance, deficiencies, or excess nutrients intake. It is classified as undernutrition or overnutrition
- Undernutrition is either stunting (low height for age), underweight (low weight for age), and wasting (low weight for height) (Kenya: Nutrition Profile, 2021).
- Proper nutrition is a crucial factor for the growth a development of children under five.

Statement of the problem

- In Kenya, malnutrition rates are a cause for concern, with high numbers in certain regions. It exposes children to various infections, which complicate the outcome
- Results from the Kenya Demographic Health Survey (2014/2015) indicate that approximately 26% of the children under five are stunted, 4% are wasted, and 11% are underweight.

Objectives

General Objective

To determine the factors associated with under-five malnutrition in Kenya using the KDHS 2014 dataset.

Specific Objectives

1. To estimate the prevalence of malnutrition among children under-five in Kenya.
2. To determine the association between bio-demographic and socio-economic factors and under-five malnutrition.
3. To fit a multivariate model to determine factors associated with under-five malnutrition.

Justification

Literature has shown wanting statistics on malnutrition that calls for action hence the purpose of this study. This study was aimed at providing awareness of the factors associated with malnutrition among children under five. the results from this study help the Government and Policy makers to make and implement policies to reduce malnutrition rates in Kenya. This in turn helps them achieve the Big 4 Agenda and certain SDG's. [a] [b]

[a] SDGs - Sustainable Development Goals
[b] SDG 2 and 3

Literature Review

- Several policies and measures have been put in place in the country to address the issue of malnutrition (GHI, 2021). Although these policies have helped counter the rising rate of malnutrition, cases are still persistent. (Kenya: Nutrition Profile, 2021)
- Socioeconomic factors are lifestyle elements and measurements of both social standing and financial viability. These include: maternal nutrition, maternal education, wealth index and place of residence. (KDHS,2015 ;Kenya Nutrition Profile, 2021)
- Bio-demographic factors are factors in which biological considerations such as evolutionary, epidemiological and genetic determinants are accentuated. Child health and survival can be affected by alterations in reproductive patterns through various mechanisms, more so through changes in; birth order, maternal age at birth or birth intervals (Sathiya Susuman et al., 2016).
- Several studies have been done on the topic, however, they have used Logistic Regression approach and Poisson Regression. These however tend to overestimate the risk of the outcome variable when the event of interest is not rare. Modified Poisson has been seen to provide a much more concise Confidence Interval based on studies conducted by Zou (Zou et.al,2004).

Methodology

- The study used the Kenya Demographic Health Survey (KDHS 2014) dataset. Specifically, the interest was with the Children's recode (KEKR72FL).
- Descriptive statistics was done to summarize the data quantitatively.
- The summary statistics also gave a general outlook of the data.
- Bivariate analysis was done using Chi-square test to examine the strengths of the relationship among variables of interest.
- For multivariate analysis, the model applied was the Modified Poisson Regression model.
- The model is defined as follows:

$$log[\pi] = \alpha + \beta Xi$$

- The Xi's are the independent variables. in this case: Wealth index,level of education,maternal age, place of residence and sex of the child.
- The KDHS 2014 dataset has a lot of variables, for the analysis, the variables of interest were kept. The dependent variables were generated and coded in binary form.
- Some other independent variables such as total number of people the household, birth order, succeeding birth interval and size of the child at birth were recoded and categorized.

Results

- From the results, 26.01% were stunted and 10.87% were underweight.
- A majority of stunted children (15.08%) and underweight children (7.2%) were from low income households (lowest and second lowest categories).
- Rural areas had a higher proportion of malnourished children than urban areas where 19.2% of children were Stunted and 8.48% were underweight.
- Males had a higher proportion of stunting (15.06%) and underweight (6.08%) as compared to females who had 10.95% stunted and 4.79% underweight.
- In terms of Region, a majority of the stunted and underweight children were from Rift Valley 8.72% and 4.52% as compared to Coast region.

The Modified Poisson Regression Model

Variable	S.IRR (95% CI)	p-value	U. IRR (95% CI)	p-value
Maternal Education				
No education	1.22 (0.90,1.67)	*0.203*	2.00 (1.13,3.51)	0.016
Primary	1.58 (1.18,2.12)	0.002	1.74 (1.00,3.03)	0.05
Secondary	1.19 (0.88,1.60)	*0.258*	1.16 (0.64,2.06)	*0.623*
Higher	Reference Category			
Type of Place of Residence				
Rural	1.88 (1.48,1.99)	0.027	1.83 (1.39,2.00)	0.002
Urban	Reference Category			
Wealth Index				
Poorest	2.25 (1.78,2.83)	<0.001	3.23 (2.16,4.83)	<0.001
Poorer	1.78 (1.41,2.24)	<0.001	2.12 (1.41,3.18)	<0.001
Middle	1.56 (1.23,1.98)	<0.001	1.86 (1.22,2.82)	0.004
Richer	1.50 (1.98,1.91)	0.001	1.59 (1.03,2.44)	0.034
Richest	Reference Category			
Size of Child at Birth				
Very large	0.80 (0.51,1.23)	*0.307*	0.43 (0.21,0.87)	0.02
Larger than average	0.86 (0.60,1.23)	*0.411*	0.52 (0.30,0.90)	0.019
Average	0.96 (0.67,1.36)	*0.807*	0.68 (0.40,1.15)	0.151
Smaller than average	1.36 (0.95,1.95)	*0.095*	1.26 (0.73,2.17)	0.402
Very small	1.61 (1.10,2.37)	0.015	1.50 (0.85,2.67)	0.164
Large	Reference Category			

The Modified Poisson Regression Model

Variable	S. IRR (95% CI)	p-value	U.IRR (95% CI)	p-value
Sex of the Child				
Male	1.37 (1.25,1.49)	<0.001	1.3623 (1.18,1.57)	<0.001
Female	Reference Category			
Maternal Age				
15-19	2.16 (1.38,3.38)	0.001	1.35(0.63,2.92)	*0.441*
20-24	2.07 (1.36,3.16)	0.001	1.38(0.73,2.60)	*0.321*
25-29	1.90 (1.26,2.87)	0.002	1.90 (1.03,3.48)	0.04
30-34	1.78 (1.18,2.68)	0.006	1.68 (0.92,3.05)	*0.089*
35-39	1.56 (1.04,2.35)	0.032	1.63 (0.91,2.91)	*0.098*
40-44	1.43 (0.94,2.21)	*0.097*	1.73 (0.95,3.17)	*0.074*
45-49	Reference Category			

[1] [2]

[1] Highlighted cells are those with p-value>0.05

[2] S.IRR & U.IRR are Stunting and Underweight Incidence Rate Ratios respectively

Discussion

- In the multivariate Modified Poisson model, wealth index, type of place of residence, sex of the child, size of the child at birth, maternal education were significantly associated with both Stunting and Underweight among children under five.
- Our results were in agreement with what is in literature and past studies.
- According to a study by Ambel in 2014, education determines the amount of health knowledge a mother has which is equally essential in child health.
- In urban areas, maternal health services such as ANC, PNC visits and health facility delivery are readily available and can equally raise community awareness and provide child immunization services and quality complementary feeding (Bado & Susuman, 2016) .
- Low income households have limited resources and thus less likely to access health facilities, expand their knowledge on access to health services and better nutritional practices that ensure children get a balanced diet (Bhagowalia et al., 2012).

[3] [4]

[3] ANC- Antenatal Care

[4] PNC - Postnatal Care

Conclusion

- From the study, it was notable that the prevalence rates of malnutrition among under-fives were still a cause for concern and need to be addressed.
- Also, socio-economic and bio-demographic characteristics such as type of place of residence, region, wealth-index , birth-order, maternal age, size of the child at birth and maternal education were found to be significant in determining a child's nutrition status.
- The study revealed that low income households, residing in the rural areas, with primary education level and lower contribute the highest cases of under-five malnutrition in Kenya.
- Addressing these factors will likely reduce the cases of malnutrition.
- The factors that influence malnutrition cases in Kenya are interrelated and thus there may be overall blanket program that can solve the problem at once.
- A more rigorous approach is looking at the problem from a multi-disciplinary perspective.

Recommendations

- A greater majority of the bio-demographic predictors that are maternal related are more personal. The most effective government intervention is civic education. From our results, we recommend the Ministry of Health in conjunction with county governments should carry out campaigns , general advocacy and advertisements .
- The region differences can be addressed by increasing the funding dedicated towards healthcare.
- There is need for further research that uses primary data where variables that were not captured are incorporated.

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